Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/20/2013
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Utilizing the Advanced Biological Research System (ABRS) on the International Space Station (ISS) to Uncover Microgravity's Impact on Root Development and Cell Wall Architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Organization Name:	NASA Kennedy Space Center		
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Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/25/2012	**End Date:**	07/24/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX12AM94G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall goal of this proposal is to uncover the underlying molecular mechanisms by which microgravity impacts cell wall architecture in the model plant Arabidopsis thaliana using the Advanced Biological Research System (ABRS) housed on the International Space Station (ISS). This proposal builds on a robust dataset from our rapid turnaround spaceflight experiments, which utilized the Biological Research in Canisters (BRIC16) hardware on the STS-131 mission. Through experiments with the BRIC hardware, we discovered that spaceflight had a profound influence on the expression of genes involved in regulating cell wall architecture. This led us to hypothesize that plants respond to microgravity in large part through the transcriptional reprogramming of genes that control cell wall remodeling. Because of the prominent root developmental phenotypes (i.e. primary root skewing and inhibition of root hair growth) that we observed in our BRIC experiments, it appears that microgravity-dependent changes in cell wall-related transcripts are predominantly manifested in roots. Furthermore, because primary root and root cell wall ultrastructural defects were most pronounced in a mutant to a vegetative actin isoform (act2), we hypothesize that microgravity predominantly affects genes that lie within signaling networks linking the actin cytoskeleton with the plant extracellular matrix. Given our successful experiments with BRIC, we are in an excellent position to effectively utilize the unique capabilities of the ABRS hardware on the ISS to test the aforementioned hypotheses. We will use a new algorithm that we developed from our BRIC16 experiments to more rigorously quantify root growth behavior of wild-type and a higher order vegetative actin mutant (act2 act8 double mutant) (aim 1). This will clarify the extent of actin involvement in specifying root orientation and cell wall remodeling in space. Through the use of monoclonal antibodies and immunocytochemistry, we will also determine how the distribution and abundance of specific cell wall components change in microgravity, and correlate such changes with root developmental responses and modifications in gene expression (aim 2). Finally, we will employ more sensitive deep sequencing technologies (e.g. RNA-Seq) to discover novel microgravity-induced transcriptional changes in plants (aim 3). Knowledge from our spaceflight studies will then be translated into corresponding gene functional studies (e.g. mutant analysis and overexpression), which we expect will lead to the discovery of new players that modulate root development on earth and in microgravity. The proposed work is therefore relevant to this NASA-NRA because it will address fundamental questions on how gravity modulates the plant's internal cytoarchitecture and the molecular processes that govern the development of its cell walls. This will be significant for NASA's mission as extrapolated in the priorities of the Fundamental Space Biology Plan for 2010-2020 because it will provide us with basic knowledge to guide genetic engineering efforts to generate plants that are better adapted to the weightless environment of spaceflight. This in turn will be beneficial for the design of large plant habitats on the ISS and future space colonies.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a deeper understanding of how microgravity influences plant development. The focus of this research will be on the development of roots and how they modify the expression of their genes to adapt to the microgravity environment of spaceflight. This research will address one of high priority recommendations of the National Research Council's 2011 decadal survey report, which is to contribute significantly to building a robust spaceflight program to analyze plant physiological responses to stimuli encountered in microgravity. Research areas depicted in this proposal namely root development and cell wall biology are significant for NASA's goal of developing large plant habitats on the International Space Station (ISS) and future space colonies. We expect that information gained from basic studies of root growth in microgravity will guide the development of plant varieties better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Furthermore, because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of its cell walls, this research will provide a deeper mechanistic understanding of the genetic networks that specify how cell walls are constructed in microgravity. In space, such knowledge will enable technologies for designing plants that can sustain optimal growth and stature despite the adverse impact that microgravity has on cell wall development. With regard to Earth benefits, understanding of molecular pathways that govern root growth and cell wall development, will benefit agricultural practices and bioenergy research on Earth.
Task Progress:	During the first year of the grant we completed the Science Verification Test (SVT) and the Payload Verification Test (PVT) both of which were conducted at the John F. Kennedy Space Center (KSC) in Cape Canaveral, Florida. The SVT was held in October, 2012 and the PVT in March, 2013. We also completed crew training with Astronaut Michael Hopkins at the Lyndon B. Johnson Space Center (JSC) in Houston, Texas in May, 2013. All three operations were successful and their completion now place us in an excellent position for launching our experiments, which are scheduled for deployment to the ISS on Space X-3 in the Fall, 2013. For both SVT and PVT, we grew Arabidopsis seedlings in six 9 cm square Perit dishes on the Advanced Biological Research System (ABRS) hardware ground unit for 7 and 14-days. RNA from 14-day old seedlings was extracted to test RNA integrity number (RIN) and yield. The data allowed us to evaluate the suitablity of the ABRS hardware growth conditions for collecting samples for aim 3 of the proposal. We found that RNA samples from ABRS-grown Arabidopsis seedlings during SVT and PVT met the minimum requirements for RIN and yield. For SVT, RIN numbers ranged from 8.0 to 8.7 (with a RIN of 10 being the highest possible value). On the other hand, RNA yield ranged from 51 ng/uL to 452 ng/uL and this should be sufficient for the proposed RNA-Seq experiments. For PVT, RIN values ranged from 7-10. One change we made during PVT was that we were able to separate shoots from the roots and still get sufficient RIN and RNA yields. We expect that this improvement will add significant value to the actual ISS-ABRS experiments because it would allow us to explore microgravity-induced gene expression changes specific to each organ type. Seed germination during SVT and PVT was excellent (almost 100% in both trials). During PVT, we refined a simple method for delaying Arabidopsis seed germination until the the imbibed seeds were exposed to white light on ABRS. This method involved exposure of planted seed to red light and maintaining Petri dishes in cold and in the dark prior to installation on ABRS. For SVT, we processed seedlings fixed in Glutaraldehyde for light microscopy to examine general root morphology and quality of labeling using selected cell wall specific monoclonal antibodies (aim 1 of this proposal). We found that antigenicity to the cell wall was adequately preserved in root cells from seedlings grown on ABRS. Although, immunocytochemistry with cell wall antibodies was successful, we noted some problems in some aspects of sample preservation. For example, primary cell walls fixed in the Kennedy Fixation Tubes (KFTs) showed smeared patterns when stained with toluidine blue dye. In optimally fixed material, primary cell walls should display a crisp bright blue and purple color. There was also degradation of cytosolic components of the root and suggests that glutaraldehyde degraded slightly during SVT. We made some slight modifications in the fixative concentration for PVT and are curently evaluating the material. We also traveled to JSC to train Astronaut, Col. Michael Hopkins on the harvest and fixation of 7- and 14-day old Arabidopsis seedlings. Col. Hopkins indicated that the harvest and fixation procedures were very clear, and should be easily implemented onboard the ISS. During this reporting period, two peer-reviewed articles were published. One article was a review paper in the tropisms issue of the American Journal of Botany. This paper critically examines how the actin cytoskeleton mediates plant gravitropism, a topic that is central to this proposal. Another paper that will be published in a 2013 plants in space special issue of the journal Plant Biology describes root morphology and cell wall ultrastructure from Arabidopsis seedlings grown on the Biological Rsearch in Canisters (BRIC). It is important to point out the latter publication in this report because our proposed experiments on ABRS builds on our BRIC results and a root orientation algorithm described in the aforementioned paper will be applied to quantify root orientation (aim 1 of this proposal).

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